Inclusive Access in People-Friendly Streets: an Agent-based Model of Mobility Constraints and Exemptions

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Declaration: LLM is used for fixing coding issues, tidying writing issues and equation formatting into Microsoft Word required formats.

1. Partial Overview Design Details (ODD) Description

1.1. Purpose and patterns

The purpose of this theoretical explanation model is to examine how vehicle restriction policy “People-Friendly Streets” (PFS) affects different accessibility outcomes for people with different mobility capacities. Therefore, this sets the research question ***How do individual exemptions under vehicle restricted “People-Friendly Streets” affect accessibility inequality between healthy and mobility constrained people?*** In this model, we will take into account two agent types and three policy settings to simulate discrepancies between groups and assess their corresponding outcomes.

The model is evaluated by three key qualitative patterns: **Pattern 1:** Mobility-constrained people to experience greater travel burden than healthy people. **Pattern 2:** Introducing vehicle restrictions (i.e. PFS) widens the accessibility gap as constrained people who tend to depend on vehicles are disproportionately affected. **Pattern 3:** Individual exemptions reduce this gap relative to the PFS, the extent is moderated by the eligible exemption uptake rate.

1.2. Entities, state variables, and scales

1.2.1. Entities

The model involved three main entities, people agents, patches and the global environment. Individuals as `healthy-people(s)` and `constrained-people(s)` moves from home (origin) patches to service (destination) patches. The simulation involved a simple abstract urban neighbourhood including road, home, service, restricted and exemption points. Under the PFS and individual-exemption scenarios, the centre vertical road becomes vehicle restricted, while including restriction point for exemption under individual-exemption.

1.2.2. State Variables

Table 1. Key agent state variables

Entity	Variable Name	Variable Type	Meaning
Healthy and constrained people	`mobility-capacity`	Numeric	Maximum capacity of burden each agent can tolerate before failing to access services
	`travel-burden`	Numeric	Accumulated burden from movement, including switching modes of transport
	`access-score`	Numeric (0-100)	An accessibility score considering burden
	`accessed?`	Boolean	Whether the agent has reached services patches
	`failed-to-access?`	Boolean	Whether the agent is failed to access services patches
	`using-vehicle?`	Boolean	Whether the agent is currently travelling by vehicle
	`switched-to-walking?`	Boolean	Whether the agent has switched mode of transport from vehicle to walking
Constrained people	`eligible-exemption?`	Boolean	Whether the agent is eligible for individual exemption

Table 2. Key global variables

Variables	Variable type	Meaning
`scenario`	Category, Interface Chooser	Policy scenario: baseline, people-friendly-streets, individual-exemption
`constrained-access-flow`	Number, Interface Slider	Percentage probability of constrained agents entering the model
`max-active-population`	Number, Interface Slider	Maximum number of people allowed in the simulation
`healthy-min-capacity`; `healthy-max-capacity`	Number, Interface Slider	Minimum, maximum mobility capacity for healthy people

`constrained-min-capacity`; `constrained-max-capacity`	Number, Interface Slider	Minimum, maximum mobility capacity for constrained agents
`exemption-uptake-rate`	Number, Interface Slider	Percentage probability of constrained people receive an exemption under `individual-exemption` scenario
`healthy-move-cost`; `constrained-move-cost` ; `vehicle-move-cost`	Number	Cost of movement per step for healthy; constrained people; vehicle. Model assumption values were inspired by Wert et al. (2013) to represent variation in movement costs across agent types and transport mode.
`mode-switch-cost`	Number	Additional cost for situations when people switches mode of transport
`access-records`	List	Accessed results records for reporters use

1.2.3. Scales

The model has a spatial extent of 33*33 grid cells, simulating an abstract urban environment. One patch is equivalent for approximately 15m, road corridors are one patch wide of approximately 15m. Distances and accessibility measures were used as a transport proxy (Tomasiello, Giannotti and Feitosa, 2020). Each tick represents one movement step for each active agent.

1.2. Process overview and scheduling

The simulation starts with setup procedure with an initial road environment, origin, destination and reset ticks. Scenario specific restrictions, maximum active population, access flow of agents, healthy and constrained people mobility capacity ranges and exemption uptake rate are set by users upon selection. Each ticks follows schedule of:

1. At each tick, new healthy or constrained agents are generated at available home patches until the number of active agents reaches `max-active-population`,

2. For each tick, active agents move one patch towards service patches, eligible exempted vehicle users may pass through exemption point patches under individual-exemption scenario. This updates travel burden, mode of transport, location and status of accessed/ failure,
3. If a vehicle user encounters a restricted road patch that is unable to enter by vehicle, the agent may switch to walking and continue moving, an additional mode-switch cost will be added to its travel burden,
4. Agent reached service patches will be updated as accessed, change to blue colour and results recorded,
5. Agent who exceeded its mobility capacity will be updated as failed to access, change to red colour and results recorded,
6. Agents that recorded failed to accessed and accessed agents will continue to move and leave the model,
7. The simulation moves by one tick

1.3. Initialisation

The model is initialised with a simplified road environment of three horizontal and three vertical roads intersecting each other. Road patches on the left of the world as home patches and road at the right as services patches. Scenario initialisation differs upon selection of policy environment of a baseline of no vehicle restrictions, a vehicle restriction corridor in the centre vertical road under non-baseline scenarios (PFS and individual-exemption). Additionally, the individual-exemption scenario has three exemption points. No agents are created during setup, they are initialised when entering the model through unoccupied home patches with capacity, mode of transport, exemption eligibility and access status variables allocated at start. Capacity will be given randomly upon user's selection range. Agents start by walking/ using a vehicle, 60% of healthy and 70% of constrained agents start as vehicle users, rest for walking.

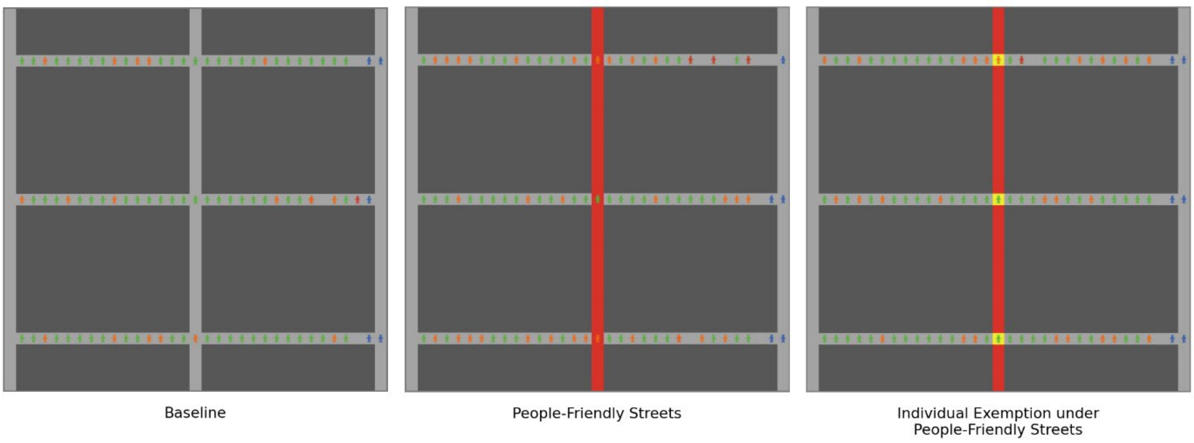

Figure 1. Model simulation under scenarios of different policy settings

1.4. Input data

This model does not use any data files from external sources to represent time-varying processes.

1.5. Submodels

When each agent moves one patch per tick (t) towards service patches. In baseline, vehicles can enter all road patches. Under PFS, vehicle users cannot enter restricted patches by vehicle and switch to walking. Under individual exemption, eligible constrained vehicle users can enter restricted patches only at exemption points. Burden accumulates as:

$$\text{travel-burden}_{i,t+1} = \text{travel-burden}_{i,t} + \text{movement-cost}_{i,t}$$

Movement cost depends on agent type (i) and its travel mode, where healthy walking as 1, mobility-constrained walking as 2, vehicle travel as 0.5 and switching transport modes as 1.5. Healthy and constrained agents are given a random capacity between `healthy-min-capacity`, `healthy-max-capacity`; `constrained-min-capacity`, `constrained-max-capacity`. It will be considered successful access if an agent reaches a service patch before exceeding capacity. Score for failed to access will be 0, while successfully accessed will be calculated as:

$$\text{access-score} = 100 \times \left(1 - \frac{\text{travel-burden}}{\text{mobility-capacity}}\right)$$

2. Situation of Model Within the Literature

Lucas (2012) discussed that the interaction of inadequate transport access and social disadvantage can lead to transport poverty through inaccessibility to essential goods and services and exclusion from planning and decision-making, which further exacerbates social exclusion and transport inequalities. Alleviating accessibility inequalities is not straightforward as transport outcomes emerge from complex behaviours and interactions within urban environments, therefore, simulating transport policy allows obtaining accessibility insights under different intervention designs and evaluating policy effects (Huang *et al.*, 2022). Agent-based modelling (ABM) has been used for exploring accessibility inequalities between groups, while is also recognised to be able to reflect mobility issues to inform future transport policy (Wise, Crooks and Batty, 2017; Tomasiello, Giannotti and Feitosa, 2020).

Existing studies largely concentrated on proposing conceptual frameworks and schemas, where simulation examining policy mechanisms for accessibility inequality for different mobility groups remains under explored. Identified by Church et al.(2000), physical exclusion is a key dimension of transport exclusion which influence people's ability to access opportunities. Therefore, the model distinguishes between different agent groups as vehicle restrictions does not affect people in the same way. Constrained agents such as having cognitive, chronic fatigue and impairments may experience higher walking burden and vehicle dependency. The model parameterises vehicle use as 60% for healthy agents and 70% for constrained agents, informed by Department for Transport (2025). Although PFS schemes are often stated with inclusive and better public health objectives, yet often fall short in creating unprecedent barriers for constrained which may lead to this group of user to "alter routes" or "avoid the area" (Lawson *et al.*, 2022). Therefore, the model takes this through mode-switch costs, road restriction designs and capacity mechanisms.

Methodologically, this follows ABM design concept of heterogenous agents, stochasticity, adaptation and emergence (Grimm *et al.*, 2020). The observed inequality emerges from heterogeneous agent capacities, stochastic vehicle and exemption usage assignment, and interaction with the restricted road environment. To conclude, the model offers a theoretical mechanism for exploring how such policies may produce uneven accessibility outcomes, and how implementation of exemption measures may mitigate these effects.

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